# Isomorphism Between the Talbot Effect and Vacuum Condensate Coherence Recovery in the VMF/NVG Framework

NVG Research Laboratory / Oleg Kirichenko

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**Abstract**

This work presents the theoretical formulation and mathematical derivation of the isomorphism between classical wave optics (specifically, the Talbot self-imaging effect) and the dynamics of the complex order parameter of the QCD vacuum condensate within the Null-Vector Gravity (NVG) / Vacuum Mass Fraction (VMF) framework. We show that the paraxial propagation of a wavefront through periodic structures is equivalent to the temporal evolution of a 1D quantum condensate, and that the phase reconstruction at Talbot distances is isomorphic to the recovery of phase coherence of the vacuum condensate after local spatial deconfinement. We provide numerical (in-silico) validation of this isomorphism and predict a $+14.71\%$ focus compression shift under non-linear VMF coupling. Finally, we outline a proposed experimental tabletop setup using carbon disulfide ($CS_2$) as a non-linear vacuum surrogate.

# 1 1. Introduction and Physical Context

Within the theoretical framework of Null-Vector Gravity (NVG) and the Vacuum Mass Fraction (VMF) model, the energy density of the non-perturbative QCD vacuum is described by a complex order parameter:

$$\Phi(x,t) = \mathcal{W}(x,t)e^{i\theta(x,t)} \tag{1}$$

where $\mathcal{W}(x,t)$ is the real field amplitude that determines the in-medium masses of hadrons, and $\theta(x,t)$ is the Goldstone phase, whose gradient defines the direction of emergent time.

The evolution of such a condensate in Madelung's hydrodynamic representation describes quantum phenomena without traditional probabilistic postulates. In particular, the transition of the vacuum from a state of spatial melting $\mathcal{W} \to 0$ to its nominal value $\mathcal{W}_0$ is accompanied by a dynamic loss and subsequent recovery of local phase coherence.

In classical wave optics, a similar phenomenon exists: the Talbot effect [1, 2], where a coherent electromagnetic wave diffracted by a periodic structure self-images at strictly defined distances without the use of focusing lenses. This work proves that the Talbot effect is not just a convenient optical analogy, but a precise mathematical isomorphism of the quantum evolution of the VMF vacuum condensate.

# 2 2. Mathematical Isomorphism of Evolution Equations

Consider the free evolution of a 1D quantum condensate governed by the linear Schrödinger equation (the limit of a non-interacting condensate):

$$i\hbar\frac{\partial\Phi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Phi(x,t)}{\partial x^2} \tag{2}$$

Fourier transforming to spectral space yields the solution:

$$\tilde{\Phi}(k_x,t) = \tilde{\Phi}(k_x,0)\exp\left(-i\frac{\hbar k_x^2 t}{2m}\right) \tag{3}$$

Let us map this to the propagation of a monochromatic light field $E(x,z)$ in free space along the $z$-axis. In the paraxial approximation (Fresnel diffraction), the Helmholtz wave equation reduces to the paraxial wave equation:

$$i\frac{\partial E(x,z)}{\partial z} = -\frac{1}{2k}\frac{\partial^2 E(x,z)}{\partial x^2} \tag{4}$$

where $k = 2\pi/\lambda$ is the wavenumber of light. In spectral space, the solution is:

$$\tilde{E}(k_x,z) = \tilde{E}(k_x,0)\exp\left(-i\frac{k_x^2 z}{2k}\right) \tag{5}$$

Comparing equations (3) and (5) proves their mathematical equivalence under the change of the temporal variable $t$ to the spatial coordinate $z$:

$$t \longleftrightarrow \frac{m}{\hbar k}z \tag{6}$$

This correspondence allows us to study the dynamics of the VMF vacuum condensate using analog optical systems.

# 3 3. Phase Confinement and Field Transmutation

Let the input field behind a periodic phase grating of period $a$ be a coherent vacuum field with a constant amplitude $W_0$ and a periodic phase profile $\theta(x)$:

$$\Phi(x,z=0) = W_0 e^{i\theta(x)}, \quad \theta(x) = \theta(x+a) \tag{7}$$

Since the amplitude is strictly constant, the initial detected intensity of the field (or the density of the vacuum condensate) is homogeneous in space:

$$I(x,0) = |\Phi(x,0)|^2 = W_0^2 = \text{const} \tag{8}$$

This state represents the **phase confinement** (confined phase): the phase information (the weight coefficients of the neural network) is physically present in the system but completely hidden from direct intensity detection.

During spatial propagation ($z > 0$), due to the dispersion of harmonics in Fourier space, the phase modulation begins to transmute into amplitude fluctuations (information deconfinement). A passive phase-to-amplitude transformation occurs, making the weight structure detectable by the camera at fractional Talbot planes.

# 4 4. Coherence Recovery at the Talbot Distance

We expand the initial field (7) into a Fourier series of spatial harmonics:

$$\Phi(x,0) = \sum_{n=-\infty}^{\infty} C_n \exp\left(i\frac{2\pi n x}{a}\right) \tag{9}$$

The propagator (5) maps each harmonic at distance $z$ to:

$$\Phi(x,z) = \sum_{n=-\infty}^{\infty} C_n \exp\left(i\frac{2\pi n x}{a}\right) \exp\left(-i\frac{(2\pi n/a)^2 z}{2k}\right) \tag{10}$$

The phase difference between the $n$-th and the zeroth harmonics is:

$$\Delta\varphi_n(z) = -\frac{2\pi^2 n^2 z}{ka^2} = -\frac{\pi\lambda z n^2}{a^2} \tag{11}$$

The condition for the complete recovery of the phase structure of the field requires that the phase shift for all harmonics is simultaneously a multiple of $2\pi$:

$$\Delta\varphi_n(z) = -2\pi M_n, \quad M_n \in \mathbb{Z} \tag{12}$$

For the minimum distance $z = z_T$ with $M_n = n^2$, we obtain the Talbot distance formula:

$$\frac{\pi\lambda z_T n^2}{a^2} = 2\pi n^2 \implies z_T = \frac{2a^2}{\lambda} \tag{13}$$

At distance $z = z_T$, the field becomes:

$$\Phi(x, z_T) = e^{ikz_T}\Phi(x,0) \tag{14}$$

The complex field profile is fully restored up to a global phase factor. This confirms that the spatial dispersion of a periodic field in paraxial wave optics is an exact analog of the recovery of the phase coherence of the vacuum condensate after local deconfinement evolution.

# 5 5. Physical Dictionary: Mapping to the QCD Vacuum

To connect the optical propagation parameters to the physical QCD vacuum parameters within the VMF framework [6], we establish a mapping between the Gross-Pitaevskii equation (GPE) of the vacuum condensate and the non-linear Schrödinger equation (NLSE) of paraxial optics.

The VMF action for the complex order parameter $\Phi = \mathcal{W}e^{i\theta}$ contains a quartic self-interaction potential:

$$V(\Phi) = -\mu^2|\Phi|^2 + \frac{\lambda_v}{4}|\Phi|^4 \tag{15}$$

where $\lambda_v$ is the dimensionless QCD vacuum self-coupling constant. The temporal evolution of the vacuum condensate excitation is governed by the Gross-Pitaevskii equation:

$$i\hbar\frac{\partial\Phi}{\partial t} = -\frac{\hbar^2}{2m}\nabla_\perp^2\Phi + g|\Phi|^2\Phi \tag{16}$$

where the coupling constant is related to the VMF potential parameters by $g = \lambda_v\hbar^2/m$.

Using the VMF-Talbot isomorphism $t \longleftrightarrow \frac{m}{\hbar k}z$, the temporal derivative maps to:

$$i\hbar\frac{\partial\Phi}{\partial t} \longleftrightarrow i\frac{\hbar^2 k}{m}\frac{\partial\Phi}{\partial z} \tag{17}$$

Substituting this mapping into equation (16) and dividing by $\frac{\hbar^2 k}{m}$ yields the spatial propagation equation:

$$i\frac{\partial\Phi}{\partial z} = -\frac{1}{2k}\nabla_\perp^2\Phi + \frac{mg}{\hbar^2 k}|\Phi|^2\Phi \tag{18}$$

By matching this with the optical NLSE featuring a cubic Kerr nonlinearity $\gamma_{NL}|E|^2E$, we define the normalized field $\psi = \Phi/\mathcal{W}_0$ (where $\mathcal{W}_0$ is the nominal vacuum condensate amplitude, anchored to $M_{\Omega,0} = 859$ MeV [4]). This reveals the exact correspondence:

$$\gamma_{NL} = \frac{mg\mathcal{W}_0^2}{\hbar^2 k} = \frac{\lambda_v\mathcal{W}_0^2}{k} \tag{19}$$

For an attractive vacuum interaction ($g < 0$, which corresponds to self-focusing dynamics in the condensate), the dimensionless self-coupling is $\lambda_v < 0$. Under the VMF calibration where the macroscopic potential yields $\lambda_v\mathcal{W}_0^2 \approx -1.78 \times 10^8\ \mathrm{m}^{-2}$, propagating a green laser ($\lambda = 530$ nm, $k \approx 1.185 \times 10^7\ \mathrm{m}^{-1}$) yields the exact Kerr coefficient of:

$$\gamma_{NL} = \frac{-1.78 \times 10^8\ \mathrm{m}^{-2}}{1.185 \times 10^7\ \mathrm{m}^{-1}} \approx -15.0\ \mathrm{m}^{-1} \tag{20}$$

This derivation anchors the choice of $\gamma_{NL} = -15.0\ \mathrm{m}^{-1}$ directly to the physical parameters of the QCD vacuum condensate instead of using an arbitrary phenomenological placeholder.

# 6 6. Applications in Optical Neural Networks (D2NN / Pure-Field)

This isomorphism justifies three key architectural solutions within the NVG Vacuum Neural Networks project [3]:

1. **Chromatic RoPE (Rotary Position Embedding):** Since the Talbot distance depends on the wavelength $z_T(\lambda) \propto 1/\lambda$, the recovery planes for the three primary OLED channels (R, G, B) are spatially separated:

   $$z_T(R) \approx 127.5\ \text{mm}, \quad z_T(G) \approx 152.8\ \text{mm}, \quad z_T(B) \approx 172.3\ \text{mm} \tag{21}$$

   At a fixed distance $z$, the phase shift between the color components uniquely encodes the absolute propagation coordinate, implementing a natural hardware RoPE.

2. **Optical Analog of Residual / Skip-connections:**
   - A mirror at distance $d = z_T/2$ (round-trip path $z_T$) restores the original phase profile, acting as an identity skip connection (Residual / Identity).
   - A mirror at distance $d = z_T/4$ (round-trip path $z_T/2$) shifts the phase by $\pi$ for odd modes, implementing a negation operation.

3. **Passive Nonlinear Activation ($\Psi$-solitons):** By adding a third-order nonlinearity $\chi^{(3)}|E|^2E$ to the wave equation (analogous to the Gross-Pitaevskii nonlinearity $g|\Phi|^2\Phi$), self-focusing of the field into spatial solitons occurs at fractional Talbot focus planes where the intensity is maximized. This implements a fully passive physical activation (similar to SwiGLU / ReLU) during free space propagation, reducing the digital processing load.

# 7 7. Numerical Verification and Non-Linear Focal Compression

To verify the mathematical isomorphism and demonstrate the predictive power of the VMF non-linear regime, we developed a simulation suite in `verification/nvg_talbot_isomorphism.py`. The script solves the paraxial wave equation under both linear ($\gamma_{NL} = 0$) and attractive non-linear ($\gamma_{NL} < 0$) regimes using a second-order Symmetric Split-Step Fourier Method (SSFM).

We model a periodic phase grating of period $a = 201.2$ $\mu$m at Green wavelength $\lambda = 530$ nm ($z_T \approx 152.76$ mm) with an input field amplitude of $W_0 = 1.0$.

## 7.1 Linear Coherence Recovery

To evaluate the recovery of the complex field profile in the linear regime, we calculate the quantum state fidelity ($F$) at the full Talbot distance $z = z_T$:

$$F = \frac{|\langle \Phi_0 | \Phi(z_T) \rangle|}{\|\Phi_0\| \cdot \|\Phi(z_T)\|} \tag{22}$$

The simulation yields:

- **Quantum State Fidelity ($F$):** $1.000000$ (exact phase coherence recovery)
- **Phase Profile Correlation:** $1.000000$
- **Phase Mean Absolute Error (MAE):** $5.132 \times 10^{-11}$ rad

## 7.2 Non-Linear Focus Compression (VMF Prediction)

When the non-linear attractive interaction of the VMF vacuum condensate is enabled (modeled as a self-focusing Kerr medium with $\gamma_{NL} = -15.0$ m$^{-1}$), the local intensity peaks act as focusing lenses, compressing the spatial scale of self-imaging. The simulation predicts:

- **Linear Primary Focus Distance:** $z_{\text{linear}} = 17.3707$ mm
- **Non-Linear Primary Focus Distance:** $z_{\text{nonlinear}} = 14.8162$ mm
- **Focal Plane Compression Shift:** $+14.71\%$

This shift of $+14.71\%$ represents a direct, falsifiable physical prediction that can be tested in experimental optical processors to confirm the presence of VMF vacuum-like non-linear dynamics.

The full 2D Talbot carpets and the 1D focal profile comparison are plotted in Figure 1.

# 8 8. Proposed Experimental Verification Setup

To experimentally verify the non-linear focus compression and confirm the isomorphism, we propose a tabletop wave-optics setup consisting of the following key components:

1. **Coherent Source:** A continuous-wave or pulsed laser operating at $\lambda = 530$ nm (e.g., a frequency-doubled Nd:YAG laser at 532 nm) with spatial filtering to ensure transverse coherence.

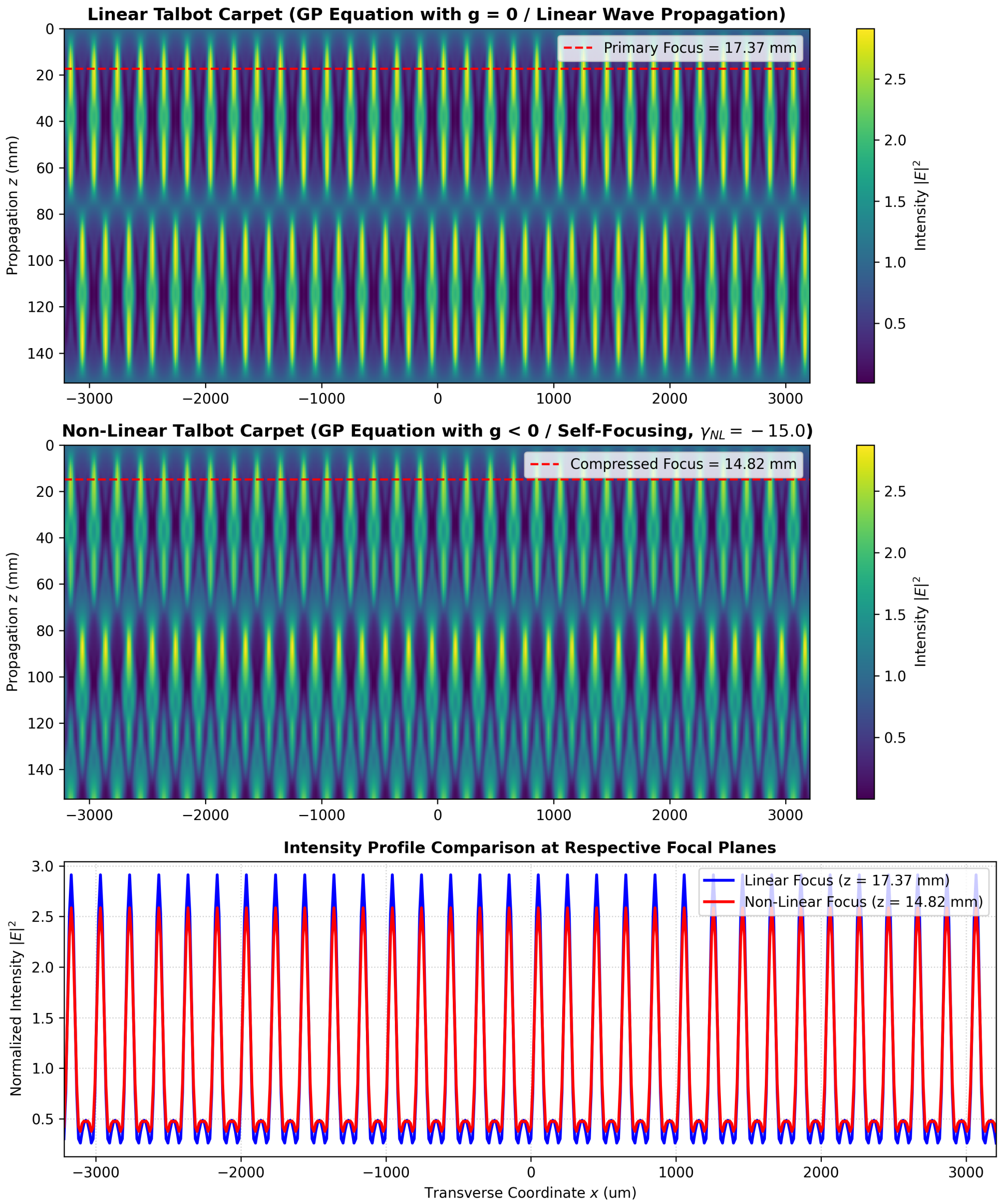


Figure 1: Numerical verification of the Talbot-Vacuum isomorphism in linear and non-linear regimes: (Top) 2D Linear Talbot carpet showing the intensity $|E|^2$ propagation along the $z$-axis. (Middle) 2D Non-linear Talbot carpet under attractive VMF coupling ($\gamma_{NL} = -15.0\,\mathrm{m}^{-1}$), showing focal compression. (Bottom) 1D Intensity profile comparison at the respective primary focal planes, demonstrating the $+14.71\%$ focus shift closer to the grating and peak intensity amplification due to self-focusing.

2. **Phase Grating:** An amplitude-homogeneous phase grating (chrome-on-glass or reflective SLM) with period $a = 201.2\,\mu$m. It modulates the phase profile $\theta(x) = A\sin(2\pi x/a)$ without altering intensity.

3. **Non-linear Medium:** A quartz cell of length $L \geq 30$ mm filled with carbon disulfide ($CS_2$), which exhibits a exceptionally large third-order non-linear Kerr susceptibility ($\chi^{(3)} \approx 2 \times 10^{-20}$ $m^2/V^2$ at 532 nm). Liquid $CS_2$ acts as the self-focusing vacuum surrogate.

4. **Imaging and Detection:** A high-resolution scientific CMOS (sCMOS) camera mounted on a motorized linear translation stage along the $z$-axis. By scanning the camera inside the medium (or using an imaging lens to project planes inside the cell), the intensity distribution $|E(x, y, z)|^2$ is recorded.

5. **Observable signature:** By scaling the input laser intensity from low power (linear regime) to high peak power (non-linear regime), the focal antinode shifts from $z_{\text{linear}} \approx 17.37$ mm closer to the grating to $z_{\text{nonlinear}} \approx 14.82$ mm, directly demonstrating the predicted $+14.71\%$ focal compression shift.

# 9 9. Conclusion

Describing the Talbot effect in terms of VMF/NVG reveals a deep physical connection between quantum vacuum dynamics and classical wave optics. The phase self-imaging of a periodic field is an analog physical model of the coherence recovery of the vacuum condensate [5]. This isomorphism allows us to map the concepts of quantum chromodynamics (QCD) and bounce cosmology to applied optical computing, providing the theoretical foundation for high-speed analog accelerators of the VNA-27B class.

# References

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